LAPORAN CASE METHOD 1

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SIB 1A

PROGRAM STUDI D4 SISTEM INFORMASI

BISNIS JURUSAN TEKNOLOGI INFORMASI

POLITEKNIK NEGERI

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Verifikasi Hasil Code:

Buku.java

```java
Buku.java src
public class Buku {
    String kodeBuku;
    String judul;
    int tahunTerbit;

    Buku(String kode, String judul, int tahun) {
        this.kodeBuku = kode;
        this.judul = judul;
        this.tahunTerbit = tahun;
    }

    void tampilBuku() {
        System.out.println("Kode: " + kodeBuku + " | Judul: " + judul + " | Tahun: " + tahunTerbit);
    }
}
```

Mahasiswa.java

```java
Mahasiswa.java src
public class Mahasiswa {    "Mahasiswa": Unknown word.
    String nim;
    String nama;
    String prodi;    "prodi": Unknown word.

    Mahasiswa(String nim, String nama, String prodi) {    "Mahasiswa": Unknown word.
        this.nim = nim;
        this.nama = nama;
        this.prodi = prodi;    "prodi": Unknown word.
    }

    void tampilMahasiswa() {    "tampil": Unknown word.
        System.out.println("NIM: " + nim + " | Nama: " + nama + " | Prodi: " + prodi);
    }
}
```

Peminjaman.java

Peminjaman.java src

```java
public class Peminjaman {    "Peminjaman": Unknown word.
    int batasPinjam = 5;    "batas": Unknown word.
    int terlambat;    "terlambat": Unknown word.
    int denda;    "denda": Unknown word.

    Peminjaman(Mahasiswa mhs, Buku buku, int lamaPinjam) {
        this.mhs = mhs;
        this.buku = buku;    "buku": Unknown word.
        this.lamaPinjam = lamaPinjam;    "Pinjam": Unknown
        hitungDenda();    "hitung": Unknown word.
    }

    void hitungDenda() {    "hitung": Unknown word.
        if (lamaPinjam > batasPinjam) {    "Pinjam": Unknow
            terlambat = lamaPinjam - batasPinjam;    "terla
            denda = terlambat * 2000;    "denda": Unknown w
        } else {
            terlambat = 0;    "terlambat": Unknown word.
            denda = 0;    "denda": Unknown word.
        }
    }

    void tampilPeminjaman() {    "tampil": Unknown word.
        System.out.println(mhs.nama + " | " + buku.judul +
                " | Lama: " + lamaPinjam +    "Pinjam": Unk
                " | Terlambat: " + terlambat +    "Terlamba
                " | Denda: " + denda);    "Denda": Unknown
    }
}
```

SistemPeminjanam.java

SistemPeminjaman.java src

You, 3 days ago | 1 author (You)

```java
import java.util.Scanner;

public class SistemPeminjaman {    "Sistem": Misspelled word.

    static Mahasiswa[] dataMhs = {    "Mahasiswa": Unknown word.
            new Mahasiswa(nim: "22001", nama: "Andi", prodi: "Teknik Informatika"),    "M
            new Mahasiswa(nim: "22002", nama: "Budi", prodi: "Teknik Informatika"),    "M
            new Mahasiswa(nim: "22003", nama: "Citra", prodi: "Sistem Informasi Bisnis")
    };

    static Buku[] dataBuku = {    "Buku": Unknown word.
            new Buku(kode: "B001", judul: "Algoritma", tahun: 2020),    "Buku": Unknown w
            new Buku(kode: "B002", judul: "Basis Data", tahun: 2019),    "Buku": Unknown
            new Buku(kode: "B003", judul: "Pemrograman", tahun: 2021),    "Buku": Unknown
            new Buku(kode: "B004", judul: "Fisika", tahun: 2024)    "Buku": Unknown word.
    };

    static Peminjaman[] dataPinjam = {    "Peminjaman": Unknown word.
            new Peminjaman(dataMhs[0], dataBuku[0], lamaPinjam: 7),    "Peminjaman": Unk
            new Peminjaman(dataMhs[1], dataBuku[1], lamaPinjam: 3),    "Peminjaman": Unk
            new Peminjaman(dataMhs[2], dataBuku[2], lamaPinjam: 10),    "Peminjaman": Un
            new Peminjaman(dataMhs[2], dataBuku[3], lamaPinjam: 6),    "Peminjaman": Unk
            new Peminjaman(dataMhs[0], dataBuku[1], lamaPinjam: 4)    "Peminjaman": Unkn
    };


    static void tampilkanMahasiswa() {    "tampilkan": Unknown word.
        System.out.println(x: "\nDaftar Mahasiswa:");    "Daftar": Unknown word.
        for (Mahasiswa m : dataMhs) {    "Mahasiswa": Unknown word.
            m.tampilMahasiswa();
        }
    }


    static void tampilkanBuku() {    "tampilkan": Unknown word.
        System.out.println(x: "\nDaftar Buku:");    "Daftar": Unknown word.
        for (Buku b : dataBuku) {    You, 3 days ago • first commit    "Buku": Unknow
            b.tampilBuku();
        }
    }

    static void tampilkanPeminjaman() {    "tampilkan": Unknown word.
        System.out.println(x: "\nData Peminjaman:");    "Peminjaman": Unknown word.
        for (Peminjaman p : dataPinjam) {    "Peminjaman": Unknown word.
            p.tampilPeminjaman();
        }
    }

```

```java
    static void urutkanBerdasarkanDenda() {    "urutkan": Unknown word.
        Peminjaman[] temp = dataPinjam.clone();    "Peminjaman": Unknown word.
        int n = temp.length;

        for (int i = 1; i < n; i++) {
            Peminjaman kunci = temp[i];    "Peminjaman": Unknown word.
            int j = i - 1;

            while (j >= 0 && temp[j].denda < kunci.denda) {    "denda": Unknov
                temp[j + 1] = temp[j];
                j--;
            }
            temp[j + 1] = kunci;    "kunci": Unknown word.
        }

        System.out.println(x: "\nSetelah diurutkan (Denda terbesar):");    "Se
        for (Peminjaman p : temp) {    "Peminjaman": Unknown word.
            p.tampilPeminjaman();
        }
    }

    static void cariBerdasarkanNIM(String nimCari) {    "cari": Unknown word.
        Peminjaman[] temp = dataPinjam.clone();    "Peminjaman": Unknown word.
        int n = temp.length;

        for (int i = 1; i < n; i++) {
            Peminjaman kunci = temp[i];    "Peminjaman": Unknown word.
            int j = i - 1;
            while (j >= 0 && temp[j].mhs.nim.compareTo(kunci.mhs.nim) > 0) {
                temp[j + 1] = temp[j];
                j--;
            }
            temp[j + 1] = kunci;    "kunci": Unknown word.
        }

        int low = 0;
        int high = n - 1;
        int found = -1;

        while (low <= high) {
            int mid = (low + high) / 2;
            int cmp = temp[mid].mhs.nim.compareTo(nimCari);    "Cari": Unknown

            if (cmp == 0) {
                found = mid;
                break;
            } else if (cmp < 0) {
                low = mid + 1;
```

```
97              } else {
98                  high = mid - 1;
99              }
100         }
101
102         if (found == -1) {
103             System.out.println("Data dengan NIM " + nimCari + " tidak ditemukan.");
104         } else {
105             int start = found;
106             int end = found;
107             while (start > 0 && temp[start - 1].mhs.nim.equals(nimCari))    "Cari":
108                 start--;
109             while (end < n - 1 && temp[end + 1].mhs.nim.equals(nimCari))    "Cari":
110                 end++;
111
112             System.out.println("Hasil pencarian NIM " + nimCari + ":");    "Hasil":
113             for (int i = start; i <= end; i++) {
114                 temp[i].tampilPeminjaman();
115             }
116         }
117     }
118
        Run | Debug
119     public static void main(String[] args) {
120         Scanner sc = new Scanner(System.in);
121         int pilihan;    "pilihan": Unknown word.
122
123         do {
124             System.out.println(x: "\n=== SISTEM PEMINJAMAN RUANG BACA JTI ===");
125             System.out.println(x: "1. Tampilkan Mahasiswa");    "Tampilkan": Unknown
126             System.out.println(x: "2. Tampilkan Buku");    "Tampilkan": Unknown word
127             System.out.println(x: "3. Tampilkan Peminjaman");    "Tampilkan": Unknow
128             System.out.println(x: "4. Urutkan Berdasarkan Denda");    "Urutkan": Unk
129             System.out.println(x: "5. Cari Berdasarkan NIM");
130             System.out.println(x: "0. Keluar");
131             System.out.print(s: "Pilih: ");
132             pilihan = sc.nextInt();
133
134             switch (pilihan) {
135                 case 1:
136                     tampilkanMahasiswa();
137                     break;
138                 case 2:
139                     tampilkanBuku();
140                     break;
141                 case 3:
142                     tampilkanPeminjaman();
143                     break;
144                 case 4:
145                     urutkanBerdasarkanDenda();
146                     break;
```

```java
            case 5:      You, 3 days ago • first commit
                System.out.print(s: "Masukkan NIM: ");
                String nimInput = sc.next();
                cariBerdasarkanNIM(nimInput);
                break;
            case 0:
                System.out.println(x: "Keluar dari program. Sampai jumpa!");
                break;
            default:
                System.out.println(x: "Pilihan tidak valid!");
        }
    } while (pilihan != 0);

    sc.close();
  }
}
```

Link Github : https://github.com/nNasq/JobsheetCM1